

Yunjin (Vivian) Yu

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ICAM-Music major and Electrical Engineering minor with hands-on electronics and embedded-hardware project experience, a strong tutoring and teaching background, and content-creation and social-media skills.

Education

University of California, San Diego

Expected Graduation: June 2027

B.A. - Interdisciplinary Computing and the Arts (Music)

- Minor: Electrical Engineering
- GPA 3.9/4.0
- Coursework: Digital Signal Processing, Machine Learning for Music, Music Programming, Circuits and Systems

UCSD Extended Studies - Certificate in Arts Management

- Digital marketing and management in the arts industry

Skills

Programming: Python (PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Matplotlib), C, C++, Java, MATLAB, Simulink

Lab Equipment: Keysight Oscilloscope, Multimeter, Power Supply, Waveform Generator, Soldering Tools

Tools & Formats: Git/GitHub, LTSpice, Onshape, PSpice, PureData

Audio/Video: AVID Pro Tools (Certified Specialist), Ableton Live, Adobe Premiere, DaVinci Resolve

Experiences

Sonic Arts R&D - Undergraduate Researcher

July 2026 - Present

Qualcomm Institute, under Professor Shahrokh Yadegari

- Develop and debug HRTF spatial audio interpolation algorithms (SOFA format) for Space3D, a real-time spatializer.
- Build SOFA preprocessing including auto-orientation, coordinate-system mapping, and ITD-preserving HRIR truncation.

Singer-Songwriter and Social Media Content Creator

Ongoing

- Release musical content under @vivianyaur on Instagram and other platforms.
- Average 2M+ monthly views with a growing 15k+ follower audience.

Projects

Daisy Seed Theremin [in progress]

July 2026 - Present

<https://github.com/vivianyunjinyu/theramen>

- Building a real-time, contactless synthesizer on Daisy Seed (STM32H750) in C++ (libDaisy/DaisySP): mapped two VL53L0X ToF sensors to continuous pitch and volume over a shared I2C bus, reassigning their identical default addresses at startup via XSHUT sequencing, alongside a 5-position waveform switch.
- Applied one-pole low-pass filtering for glide and decoupled the interrupt-driven audio callback from sensor polling.

Vocoduh - FFT Channel Vocoder

May 2026

<https://github.com/vivianyunjinyu/FFT-Vocoder-Plugin>

- Built a real-time FFT channel vocoder plugin (C++/JUCE) with STFT analysis/resynthesis engine, sidechain carrier routing, and a custom animated GUI with reported latency compensation and parameter handling via AudioProcessorValueTreeState.
- Implemented the vocoder DSP from scratch: overlap-add STFT with Kaiser windowing, mel-scale band grouping, per-band attack/release envelope smoothing, carrier spectral whitening, and frequency-weighted noise injection for unvoiced-consonant intelligibility.